

Advanced Statistics in

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Before we start with the new stuff...

- **Can you hear me OK?**
- **Homework**
 - A) too little, B) too much, C) just right?
 - A) too easy, B) too difficult, C) just right?
 - Peer-grading worked for everybody?
- **Questions/suggestions/comments/...?**

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Overview

- **Part I: Data management**
 - Leftovers from last week plus new and last part
- **Part II: Graphs and figures with the package `ggplot2`**
 - simple: `qplot()`
 - advanced: `ggplot()`
- **Homework**

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Selecting parts of a data frame

We might want to create a smaller data frame, that consists only some of the observations (= rows) or only some of the variables (= columns), or both

✓ `subset()`

✓ `==`, `!=`

✓ `&`, `|`

• `droplevels()`

• `which()`

plus

• finding and removing missing values (NA)

• recoding variables using `car::recode()`

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droplevels()

This command removes all unused factor levels

Best to **ALWAYS** use it when you select parts of a data frame with commands like `subset()`, `which()`, etc!

```
r_1_melt_f <- subset(r_1_melt, f_gender == "female")
r_1_melt_f$f_gender
```

```
[1] female female female female female female female female female
[10] female female female female female female female female female
[19] female female female female female female female female female
[28] female female female
```

```
Levels: female male
```

```
r_1_melt_f <- droplevels(subset(r_1_melt, f_gender == "female"))
```

```
[1] female female female female female female female female female ...
Levels: female
```

I am serious, ALWAYS use it when you use a command like `subset()`, `which()`, etc!

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which()

```
a <- c(49, 63, 49, 100, 20)
```

```
which(a == 49)
```

```
1, 3
```

```
which(a != 100)
```

```
1, 2, 3, 5
```

```
which(a > 60)
```

```
2, 4
```

What is this doing?

```
r_2 <- droplevels(r_1_melt[which(r_1_melt$f_gender == "male"),])
```

BTW, if you don't use `which()`, for example:

```
a == 49
```

```
gives you the answer: TRUE FALSE TRUE FALSE FALSE
```

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Missing values: NA

- How to find them
- How to remove them

```
t_1 <- read.csv("ExampleData_RT_with_Missings.csv")
```

The file is on Brightspace → Week 3

	pp_code	age	gender	RT
1	pp_1	10	1	605.5613
2	pp_2	11	2	NA
3	pp_3	NA	NA	137.9770
4	pp_4	16	2	237.3711
5	pp_5	20	2	113.9930
6	pp_6	19	1	NA
7	pp_7	7	2	788.3686
8	pp_8	15	1	129.6224
9	pp_9	20	2	454.7234
10	pp_10	7	1	265.9552

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Missing values

- First, it is important to know that they are there!

which() with == NA or something like this? Nope!

- summary()
- stat.desc() from package pastecs
- is.na()

```
summary(t_1)
```

pp_code	age	gender	RT
pp_1 :1	Min. : 7.00	Min. :1.000	Min. :114.0
pp_10 :1	1st Qu.:10.00	1st Qu.:1.000	1st Qu.:135.9
pp_2 :1	Median :15.00	Median :2.000	Median :251.7
pp_3 :1	Mean :13.89	Mean :1.556	Mean :341.7
pp_4 :1	3rd Qu.:19.00	3rd Qu.:2.000	3rd Qu.:492.4
pp_5 :1	Max. :20.00	Max. :2.000	Max. :788.4
(Other):4	NA's :1	NA's :1	NA's :2

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```
install.packages("pastecs")
library(pastecs)

stat.desc(t_1, desc = FALSE)
```

	pp_code	age	gender	RT
nbr.val	NA	9	9	8.0000
nbr.null	NA	0	0	0.0000
nbr.na	NA	1	1	2.0000
min	NA	7	1	113.9930
max	NA	20	2	788.3686
range	NA	13	1	674.3756
sum	NA	125	14	2733.5720

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stat.desc() from package pastecs

```
stat.desc(myvariable, norm = TRUE)
```

norm = TRUE tells stat.desc to compute things related to normal distribution characteristics

NOTE

- Output states skew and kurtosis, as well as skew and kurtosis divided by 2 SEs (skew.2SE and kurt.2SE)
- Thus: if the absolute value is larger than 1, the skew (or kurtosis) is significant at approx. $p < .05$
(BUT: significance tests don't make much sense here...)
- And much more: how many NA values, min, max, Shapiro Wilk test of normality, ...
→ ?stat.desc

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```
is.na(t_1)
```

```
      pp_code  age gender  RT
[1,]  FALSE FALSE  FALSE FALSE
[2,]  FALSE FALSE  FALSE  TRUE
[3,]  FALSE  TRUE   TRUE  FALSE
[4,]  FALSE FALSE  FALSE FALSE
[5,]  FALSE FALSE  FALSE FALSE
[6,]  FALSE FALSE  FALSE  TRUE
[7,]  FALSE FALSE  FALSE FALSE
[8,]  FALSE FALSE  FALSE FALSE
[9,]  FALSE FALSE  FALSE FALSE
[10,] FALSE FALSE  FALSE FALSE
```

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```
is.na(t_1$pp_code)
```

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

```
unique(is.na(t_1$pp_code))
```

```
[1] FALSE
```

```
is.na(t_1$RT)
```

```
[1] FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE
```

```
unique(is.na(t_1$RT))
```

```
[1] FALSE  TRUE
```

If we want to know which ones are NA

```
which(is.na(t_1$RT) == TRUE)
```

```
[1] 2 6
```

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If we want to create a data frame from which the rows with NA in the variable RT have been removed

```
t_2 <- droplevels(t_1[-which(is.na(t_1$RT) == TRUE),])
```

or

```
t_2 <- droplevels(t_1[which(is.na(t_1$RT) == FALSE),])
```

	pp_code	age	gender	RT
1	pp_1	10	1	605.5613
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If we want a data frame without ANY NAs

```
t_2 <- droplevels(na.omit(t_1))
```

	pp_code	age	gender	RT
1	pp_1	10	1	605.5613
4	pp_4	16	2	237.3711
5	pp_5	20	2	113.9930
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also useful: **complete.cases()**

```
complete.cases(t_1)
```

```
[1] TRUE FALSE FALSE TRUE TRUE FALSE TRUE TRUE TRUE TRUE
```

```
t_2 <- droplevels(t_1[complete.cases(t_1) == TRUE,])
```

```
t_2 <- droplevels(t_1[complete.cases(t_1),])
```

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recode()

- `install.packages("car")`
- `library(car)`
- It recodes stuff (car has many other functionalities, mainly for regression...)

t_1\$gender is coded with 1 and 2

Let's create a new variable (a factor) in which 1 is replaced with "male" and 2 with "female"

```
t_1$f_gender <- recode(t_1$gender, "1 = 'male'; 2 = 'female'")
```

```
t_1$f_gender <- as.factor(t_1$f_gender)
```

Everything in one single command

```
t_1$f_gender <- as.factor(recode(t_1$gender, "1 = 'male'; 2 = 'female'"))
```

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t_1\$age is a continuous variable

Let's create a factor f_agegroup with 3 categories

- "child" up to 12 years
- "teenager" 13 to 19 years
- "adult" 20 years and older

```
t_1$f_agegroup <- as.factor(recode(t_1$age, "0:12 = 'child'; 13:19 = 'teenager'; 20:99 = 'adult'"))
```

If we want another factor just indicating "teenager" versus "noteenager"

```
t_1$f_TeenagerOrNot <- as.factor(recode(t_1$f_agegroup, "'child' = 'noteenager'; 'teenager' = 'teenager'; 'adult' = 'noteenager'"))
```

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recode() is very handy

- if you have 99999 entries indicating missing values (from SPSS) that should be replaced by NA
- if you want to remove outliers, i.e., replace them by NA or some other value
- ...

Let's remove outliers in t_1\$RT

Let's say for this specific task, RTs below 50 msec and RTs above 1000 msec are treated as outliers. Let's create a new variables, in which such outliers are turned into NA values.

```
t_1$RT_noOutliers <- recode(t_1$RT, "lo:49 = NA; 1001:hi = NA")
```

→ more details/variants in example R script

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